Math 220: Mathematical Reasoning and Proof
Instructor: David Zureick-Brown ("DZB")

All Solutions

Last updated: November 6, 2026

1 (September 4): Introduction to course. First proof 2

2 (September 11): Logic. "Direct" proofs 3

3 (September 18): More "Direct" proofs; proof by cases 7

4 (September 25): Proof by contradiction 10

5 (October 2): Induction 12

6 (October 16): Set theory. Basic operations. Proofs with sets 15

7 (October 23): More sets. DeMorgans laws. Cartesian Products. Power sets 17

8 (October 30): Introduction to functions; images and surjectivity 21

9 (November 6): Inverse Image (or Preimage) 24

Solutions 1: Introduction to course. First proof

Solutions to appear after the assignment is due.

1. Using the definitions of 'divides" and 'even", explain why 0 is even. Explain why 7 divides 0, but 0 does not divide 7.

   Solution. Recall that an integer a is *even* if there exists an integer k such that $a = 2k$. Since $0 = 2 \cdot 0$ and 0 is an integer, it follows that 0 is even.

   Recall also that, for integers a and b, we say that *a divides b* if there exists an integer k such that $b = ak$. Since $0 = 7 \cdot 0$, there exists an integer k, namely $k = 0$, such that $0 = 7k$. Therefore, $7 \mid 0$.

   On the other hand, if 0 divided 7, then there would be an integer k such that $7 = 0 \cdot k$. But $0 \cdot k = 0$ for every integer k, so this would imply that $7 = 0$, which is false. Therefore, $0 \nmid 7$.

2. Suppose that n is an even integer, and let m be any integer. Prove that nm is even.

   Solution. Since n is even, there exists an integer k such that $n = 2k$. Multiplying by m, we obtain

   $$nm = (2k)m = 2(km).$$

   Because k and m are integers, their product km is an integer. Thus nm is twice an integer, and we conclude that nm is even.

3. Suppose that n and m are odd integers. Prove that $n + m$ is an even integer.

   Solution. Since n and m are odd, there exist integers a and b such that $n = 2a + 1$ and $m = 2b + 1$. Therefore,

   $$n + m = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

   Since a and b are integers, $a + b + 1$ is an integer. Thus $n + m$ is twice an integer, and we conclude that $n + m$ is even.

Solutions 2: Logic. "Direct" proofs

1. Write the negation of each of the following statements.

   Solution.

   (a) All triangles are isosceles.
   Negation: There exists a triangle that is not isosceles.

   (b) Every door in the building was locked.
   Negation: There exists a door in the building that was not locked.

   (c) Some even numbers are multiples of three.
   Negation: No even number is a multiple of three.
   Equivalently: Every even number is not a multiple of three.

   (d) Every real number is less than 100.
   Negation: There exists a real number that is greater than or equal to 100.

   (e) Every integer is positive or negative.
   Negation: There exists an integer that is neither positive nor negative.

   (f) If f is a polynomial function, then f is continuous at 0.
   Negation: f is a polynomial function and f is not continuous at 0.

   (g) If $x^2 > 0$, then $x > 0$.
   Negation:
   $$x^2 > 0 \quad \text{and} \quad x \leq 0.$$

   (h) There exists a $y \in \mathbf{R}$ such that $xy = 1$.
   Negation: For every $y \in \mathbf{R}$,
   $$xy \neq 1.$$

   (i) $(2 > 1)$ and $(\forall x, x^2 > 0)$.
   Negation:
   $$(2 \leq 1) \quad \text{or} \quad (\exists x \text{ such that } x^2 \leq 0).$$

   (j) $\forall \epsilon > 0$, $\exists \delta > 0$ such that if $|x| < \delta$, then $|f(x)| < \epsilon$.
   Negation: There exists an $\epsilon > 0$ such that for every $\delta > 0$, $|x| < \delta$ and $|f(x)| \geq \epsilon$.

   The following is also correct:

   There exists an $\epsilon > 0$ such that for every $\delta > 0$, there exists an x such that

   $$|x| < \delta \qquad \text{and} \qquad |f(x)| \geq \epsilon.$$

2. Write the converse, contrapositive, and negation of each of the following implications.

   Solution.

   (a) If a quadrilateral is a rectangle, then it has two pairs of parallel sides.
   Converse: If a quadrilateral has two pairs of parallel sides, then it is a rectangle.
   Contrapositive: If a quadrilateral does not have two pairs of parallel sides, then it is not a rectangle.
   Negation: There exists a quadrilateral that is a rectangle but does not have two pairs of parallel sides.

(b)

$$(P \wedge \neg Q) \Rightarrow R.$$

Converse:

$$R \Rightarrow (P \wedge \neg Q).$$

Contrapositive:

$$\neg R \Rightarrow \neg(P \wedge \neg Q).$$

Equivalently,

$$\neg R \Rightarrow (\neg P \vee Q).$$

Negation:

$$(P \wedge \neg Q) \wedge \neg R.$$

(c)

$$P \Rightarrow (R \Rightarrow \forall x,\ Q(x)).$$

Converse:

$$(R \Rightarrow \forall x,\ Q(x)) \Rightarrow P.$$

Contrapositive:

$$\neg(R \Rightarrow \forall x,\ Q(x)) \Rightarrow \neg P.$$

Equivalently,

$$(R \wedge \exists x \text{ such that } \neg Q(x)) \Rightarrow \neg P.$$

Negation:

$$P \wedge \neg(R \Rightarrow \forall x,\ Q(x)).$$

Equivalently,

$$P \wedge R \wedge \exists x \text{ such that } \neg Q(x).$$

3. Let P and Q be statements. Write the truth table for

   (a) $(\neg P) \vee Q$

   (b) $((\neg P \Rightarrow Q) \wedge \neg Q) \Rightarrow P$

Solution.

(a)

P	Q	$\neg P$	$(\neg P) \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

(b)

P	Q	$\neg P$	$\neg P \Rightarrow Q$	$(\neg P \Rightarrow Q) \wedge \neg Q$	$((\neg P \Rightarrow Q) \wedge \neg Q) \Rightarrow P$
T	T	F	T	F	T
T	F	F	T	T	T
F	T	T	T	F	T
F	F	T	F	F	T

4. Are the statements $(P \vee Q) \wedge R$ and $P \vee (Q \wedge R)$ equivalent? If so, give a proof. If not, explain why by giving a counterexample.

   Solution.

   They are not equivalent. For example, suppose $P = T$, $Q = F$, $R = F$. Then $(P \vee Q) \wedge R = (T \vee F) \wedge F = T \wedge F = F$, while $P \vee (Q \wedge R) = T \vee (F \wedge F) = T$. Thus the two statements can have different truth values, so they are not equivalent.

5. Let P and Q be statements.

   (a) Prove that $\neg(P \Rightarrow Q)$ is equivalent to $P \wedge \neg Q$.

   Solution.

   We can prove this using a truth table:

P	Q	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	F	F

   The last two columns are identical. Therefore, $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$ are equivalent.

   (b) Prove that $\neg(P \Rightarrow Q)$ is *not* equivalent to $\neg P \wedge Q$.

   Solution.

   It is enough to find a choice of P and Q for which the two statements have different truth values.
   Suppose $P = F$ and $Q = T$. Then $P \Rightarrow Q$ is true, so $\neg(P \Rightarrow Q) = F$. On the other hand, $\neg P \wedge Q = T \wedge T = T$. Thus the two statements are not equivalent.

   (c) Give an example of statements P and Q such that $\neg P \Rightarrow \neg Q$ is true and $\neg(P \Rightarrow Q)$ is false.

   Solution.

   For example, let

   $$P = \text{“}2 > 1\text{”} \qquad \text{and} \qquad Q = \text{“}3 > 2\text{”}.$$

   Both P and Q are true.
   Thus $\neg P$ is false, so $\neg P \Rightarrow \neg Q$ is true.
   Also, $P \Rightarrow Q$ is true, and hence $\neg(P \Rightarrow Q)$ is false.

6. Suppose that n is an odd integer. Prove that n^2 is an odd integer.

   Solution.

   Since n is odd, there exists an integer k such that $n = 2k + 1$. Therefore,

   $$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

   Since k is an integer, $2k^2 + 2k$ is an integer. Thus n^2 can be written in the form $2\ell + 1$ for some integer ℓ. Therefore n^2 is odd.

7. Prove that if n^2 is even, then n is even.

   Solution.

   We prove the statement by contraposition.

   The contrapositive is:

If n is not even, then n^2 is not even.

Suppose that n is not even. Since n is an integer, n must therefore be odd. By the previous problem, if n is odd, then n^2 is odd. In particular, n^2 is not even.

Thus the contrapositive is true. Therefore the original statement is true: if n^2 is even, then n is even.

8. Prove that m is even if and only if $m^2 + 6$ is even.

Solution.

We prove both directions.

First suppose that m is even. Then there exists an integer k such that $m = 2k$. Therefore $m^2+6 = (2k)^2+6 = 4k^2 + 6 = 2(2k^2 + 3)$. Since $2k^2 + 3$ is an integer, $m^2 + 6$ is even.

Conversely, suppose that $m^2 + 6$ is even. Then there exists an integer k such that $m^2 + 6 = 2k$. Then $m^2 = 2k - 6 = 2(k - 6)$. Since $k - 6$ is an integer, m^2 is even. By the previous homework problem, we conclude that m is even.

Solutions 3: More "Direct" proofs; proof by cases

1. Prove that for all $a \in \mathbb{Z}$ and for $n \in \mathbb{Z}_{\geq 0}$, $a - 1$ divides $a^n - 1$.

   Solution. If $n = 0$, then $a^n - 1 = 1 - 1 = 0 = (a - 1) \cdot 0$, so $a - 1 \mid a^n - 1$.

   Now suppose $n \geq 1$. We use the factorization

   $$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1).$$

   The expression in the second set of parentheses is an integer, so by the definition of divisibility, $a-1 \mid a^n-1$.

2. Suppose that $a \mid b$. Prove that for all $n \in \mathbb{Z}_{>0}$, $a^n \mid b^n$.

   Solution. Since $a \mid b$, there is an integer k such that $b = ak$. Raising both sides to the nth power gives

   $$b^n = (ak)^n = a^n k^n.$$

   Since k^n is an integer, this shows that $a^n \mid b^n$.

3. Suppose that there exists an integer $n \in \mathbb{Z}_{>0}$ such that $a \mid b^n$. Is it true that $a \mid b$? Prove or disprove your answer.

   Solution. The statement is **false**. For example, take $a = 4$, $b = 2$, and $n = 2$. Then $b^n = 2^2 = 4$, so $4 \mid 2^2$. However, $4 \nmid 2$. Thus it is possible for $a \mid b^n$ even though $a \nmid b$.

4. Prove that for all integers n, n and $n + 1$ have no common divisors other than ± 1.

   Solution. Suppose that d is a common divisor of n and $n + 1$. Then there are integers r and s such that $n = dr$ and $n + 1 = ds$. Subtracting these equations gives $1 = (n + 1) - n = d(s - r)$. Thus $d \mid 1$. The only integer divisors of 1 are 1 and -1, so $d = \pm 1$. Therefore n and $n + 1$ have no common divisors other than ± 1.

5. Prove that the product of three consecutive integers is divisible by 6.

   Solution. Let the three consecutive integers be n, $n + 1$, and $n + 2$. Among any two consecutive integers, one is even. Therefore at least one of these three integers is even, so $2 \mid n(n + 1)(n + 2)$.

   Also, every integer has remainder 0, 1, or 2 when divided by 3. Thus among three consecutive integers, exactly one is divisible by 3. Hence $3 \mid n(n + 1)(n + 2)$.

   Since 2 and 3 are relatively prime, divisibility by both 2 and 3 implies divisibility by 6. Therefore $6 \mid n(n + 1)(n + 2)$.

6. Show that for all integers a and b, $a^2b^2(a^2 - b^2)$ is divisible by 12.

   Solution. We will show separately that the expression is divisible by 4 and by 3.

   First consider divisibility by 4. If a is even, then a^2 is divisible by 4, so the entire product is divisible by 4. The same argument works if b is even.

   The only remaining case is when both a and b are odd. Then both $a - b$ and $a + b$ are even. Since

   $$a^2 - b^2 = (a - b)(a + b),$$

   the factor $a^2 - b^2$ is divisible by 4. Thus in every case, $4 \mid a^2b^2(a^2 - b^2)$.

   Now consider divisibility by 3. If $3 \mid a$ or $3 \mid b$, then $3 \mid a^2b^2$, so the whole expression is divisible by 3.

Suppose instead that neither a nor b is divisible by 3. Then each of a and b has the form $3k+1$ or $3k-1$ for some integer k. In either case, its square has remainder 1 upon division by 3. Thus we can write $a^2 = 3r+1$ and $b^2 = 3s+1$ for some integers r, s. Therefore

$$a^2 - b^2 = (3r+1) - (3s+1) = 3(r-s),$$

so $3 \mid a^2 - b^2$. Hence in every case, $3 \mid a^2b^2(a^2-b^2)$.

Since 3 and 4 are relatively prime, divisibility by both 3 and 4 implies divisibility by 12. Therefore $12 \mid a^2b^2(a^2-b^2)$.

7. Finish the following proofs from class.

   (a) In class, when we proved that $|xy| = |x|\,|y|$, we only did the cases when $x \geq 0$. Prove that $|xy| = |x|\,|y|$ when $x < 0$.

   Solution. Assume $x < 0$. There are two cases.

   If $y \geq 0$, then $xy \leq 0$. Therefore $|xy| = -xy = (-x)y = |x|\,|y|$.

   If $y < 0$, then $xy > 0$. Therefore $|xy| = xy = (-x)(-y) = |x|\,|y|$.

   Thus in both cases, $|xy| = |x|\,|y|$.

   (b) In class, when we proved that $|x+y| \leq |x| + |y|$, we only did the cases when $x, y \geq 0$, when $x, y < 0$, and when $x \geq 0$, $y < 0$. Prove that $|x+y| \leq |x| + |y|$ when $x < 0$ and $y \geq 0$.

   Solution. Assume $x < 0$ and $y \geq 0$. Then $|x| = -x$ and $|y| = y$. There are two cases depending on the sign of $x + y$.

   If $x + y \geq 0$, then $|x+y| = x + y$. Since $x < 0$, we have $x \leq -x$, and so

   $$|x+y| = x + y \leq -x + y = |x| + |y|.$$

   If $x + y < 0$, then $|x+y| = -(x+y) = -x - y$. Since $y \geq 0$, we have $-y \leq y$, and therefore

   $$|x+y| = -x - y \leq -x + y = |x| + |y|.$$

   Thus in either case, $|x+y| \leq |x| + |y|$.

8. Consider the following statements about inequalities.

   (a) Prove that $|a-b| < c$ if and only if $b - c < a < b + c$.

   Solution. We prove both directions.

   First suppose that $|a-b| < c$. Since $|a-b| \geq 0$, this implies $c > 0$.

   If $a - b \geq 0$, then $|a-b| = a - b < c$, so $a < b + c$. Also $a \geq b$, and since $c > 0$, we have $a \geq b > b - c$. Thus $b - c < a < b + c$.

   If $a - b < 0$, then $|a-b| = b - a < c$, so $b - c < a$. Also $a < b$, and since $c > 0$, we have $a < b < b + c$. Again, $b - c < a < b + c$.

   Conversely, suppose that $b - c < a < b + c$. Because a number a lies strictly between $b - c$ and $b + c$, we must have $c > 0$.

   If $a - b \geq 0$, then $|a-b| = a - b$. From $a < b + c$, we get $a - b < c$, and hence $|a-b| < c$.

   If $a - b < 0$, then $|a-b| = b - a$. From $b - c < a$, we get $b - a < c$, and hence $|a-b| < c$.

   Therefore $|a-b| < c$ if and only if $b - c < a < b + c$.

   (b) Prove or disprove: if $a - b < c$, then $|a-b| < c$.

   Solution. This statement is **false**. Take $a = 0$, $b = 2$, and $c = 1$. Then $a - b = -2 < 1 = c$, so the hypothesis is true. But $|a-b| = |-2| = 2 \not< 1$. Thus the conclusion is false.

(c) Prove or disprove: if $|a-b| < c$, then $a - b < c$.

Solution. This statement is **true**. By part (a), $|a-b| < c$ implies $b - c < a < b + c$. In particular, $a < b + c$. Subtracting b from both sides gives $a - b < c$.

(d) Prove that $\big||x| - |y|\big| \leq |x + y|$.

Solution. By the triangle inequality,

$$|x| = |(x + y) - y| \leq |x + y| + |y|.$$

Subtracting $|y|$ gives $|x| - |y| \leq |x + y|$. Similarly,

$$|y| = |(x + y) - x| \leq |x + y| + |x|,$$

so $|y| - |x| \leq |x + y|$.

Now there are two cases. If $|x| - |y| \geq 0$, then $\big||x| - |y|\big| = |x| - |y| \leq |x + y|$. If $|x| - |y| < 0$, then $\big||x| - |y|\big| = |y| - |x| \leq |x + y|$.

Therefore $\boxed{\big||x| - |y|\big| \leq |x + y|}$.

Solutions 4: Proof by contradiction

1. Prove that there do not exist integers a and b such that $21a + 30b = 1$.

   Solution. Suppose, for contradiction, that there were integers a and b satisfying $21a + 30b = 1$. Both $21a$ and $30b$ are divisible by 3, so their sum is divisible by 3. Thus $3 \mid (21a + 30b)$. But $21a + 30b = 1$, so this would imply that $3 \mid 1$, which is impossible.

   Therefore there are no integers a, b such that $21a + 30b = 1$.

2. Find all positive integers n such that $n^2 - 1$ is prime. Prove that your answer is correct.

   Solution. We factor $n^2 - 1 = (n - 1)(n + 1)$.

   If $n = 1$, then $n^2 - 1 = 0$, which is not prime. If $n = 2$, then $n^2 - 1 = 4 - 1 = 3$, which is prime.

   Now suppose that $n \geq 3$. Then $n - 1 \geq 2$ and $n + 1 \geq 4$. Thus $n^2 - 1$ is a product of two integers greater than 1, so it is not prime.

   Therefore the only positive integer n for which $n^2 - 1$ is prime is $\boxed{n = 2}$.

3. Prove that if x is an integer, then $x^2 + 2$ is not divisible by 4.

   Solution. There are two cases.

   If x is even, then $x = 2k$ for some integer k. Hence

   $$x^2 + 2 = (2k)^2 + 2 = 4k^2 + 2.$$

   If this were divisible by 4, then subtracting $4k^2$ would imply that 2 is divisible by 4, which is false.

   If x is odd, then $x = 2k + 1$ for some integer k. Then

   $$x^2 + 2 = (2k + 1)^2 + 2 = 4k^2 + 4k + 3 = 4(k^2 + k) + 3.$$

   If this were divisible by 4, then subtracting $4(k^2 + k)$ would imply that 3 is divisible by 4, which is false.

   Thus in either case, $\boxed{4 \nmid x^2 + 2}$.

4. Prove that there are no integer solutions to the equation $x^2 - y^2 = 2$.

   Solution. Suppose, for contradiction, that integers x and y satisfy $x^2 - y^2 = 2$.

   If x and y have the same parity, then either both are even or both are odd. If both are even, then both x^2 and y^2 are divisible by 4, so $x^2 - y^2$ is divisible by 4 and cannot equal 2.

   If both are odd, write $x = 2r + 1$ and $y = 2s + 1$. Then

   $$x^2 - y^2 = (4r^2 + 4r + 1) - (4s^2 + 4s + 1),$$

   which is divisible by 4, so it cannot equal 2.

   If x and y have different parity, then one of x^2, y^2 is even and the other is odd. Therefore $x^2 - y^2$ is odd, so it cannot equal 2.

   All possibilities lead to a contradiction. Therefore there are no integer solutions.

5. Let a, b, c be integers satisfying $a^2 + b^2 = c^2$. Prove that at least one of a, b, or c is even.

   Solution. Suppose, for contradiction, that none of a, b, c is even. Then all three are odd.

   The square of an odd integer is odd, so a^2, b^2, and c^2 are all odd. But $a^2 + b^2$ is the sum of two odd integers, and is therefore even, while c^2 is odd. Thus $a^2 + b^2 \neq c^2$, contradicting the hypothesis.

   Therefore at least one of a, b, c must be even.

6. Suppose that x is a real number such that $0 \leq x \leq \pi/2$. Prove that $\sin x + \cos x \geq 1$.

   Solution. Because $0 \leq x \leq \pi/2$, both $\sin x$ and $\cos x$ are nonnegative, so $\sin x + \cos x \geq 0$.

   Now

   $$\begin{aligned}(\sin x + \cos x)^2 &= (\sin x)^2 + 2 \sin x \cos x + (\cos x)^2 \\ &= 1 + 2 \sin x \cos x.\end{aligned}$$

   Since $\sin x \geq 0$ and $\cos x \geq 0$, we have $2 \sin x \cos x \geq 0$. Hence $(\sin x + \cos x)^2 \geq 1$. Since $\sin x + \cos x$ is nonnegative, it follows that $\boxed{\sin x + \cos x \geq 1}$.

7. Prove that $2^{1/3}$ is irrational.

   Solution. Suppose, for contradiction, that $2^{1/3}$ is rational. Then we can write $2^{1/3} = p/q$, where p and q are integers, $q \neq 0$, and p and q have no common divisor other than ± 1.

   Cubing both sides gives

   $$2 = \frac{p^3}{q^3},$$

   so $p^3 = 2q^3$. Thus p^3 is even. If p were odd, then p^3 would be odd, so p must be even. Write $p = 2k$. Substituting gives

   $$(2k)^3 = 2q^3,$$

   so $8k^3 = 2q^3$, and therefore $q^3 = 4k^3$. Thus q^3 is even, so q is even as well.

   Therefore both p and q are even, so they have a common divisor 2. This contradicts our choice of p/q in lowest terms.

   Hence $\boxed{2^{1/3} \text{ is irrational.}}$

8. Suppose that a and n are integers that are both at least 2, and suppose that $a^n - 1$ is prime.

   (a) Prove that $a = 2$.

   Solution. We use the factorization

   $$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1).$$

   Because $a \geq 2$ and $n \geq 2$, the second factor is greater than 1.

   Since $a^n - 1$ is prime, it cannot be written as a product of two positive integers that are both greater than 1. Therefore the first factor must equal 1, so $a - 1 = 1$. Hence $\boxed{a = 2}$.

   (b) Prove that n is prime.

   Solution. Suppose, for contradiction, that n is not prime. Since $n \geq 2$, this means that n is composite, so $n = rs$ for some integers $r, s > 1$.

   Then

   $$a^n - 1 = a^{rs} - 1 = (a^r)^s - 1 = (a^r - 1)\left(a^{r(s-1)} + a^{r(s-2)} + \cdots + a^r + 1\right).$$

   Because $a \geq 2$ and $r, s > 1$, both factors are greater than 1. Thus $a^n - 1$ is composite, contradicting the assumption that it is prime.

   Therefore $\boxed{n \text{ is prime.}}$

Solutions 5: Induction

1. Prove that for every positive integer n,

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

Solution.

We prove the statement by induction on n.

Base case: Let $n = 1$. Then $1^3 = 1$ and $\frac{1^2(1+1)^2}{4} = 1$. Thus the statement is true when $n = 1$.

Inductive step: Suppose that for some positive integer n, $1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$. We must prove that $1^3 + 2^3 + \cdots + n^3 + (n+1)^3 = \frac{(n+1)^2(n+2)^2}{4}$. Using the inductive hypothesis,

$$\begin{aligned} 1^3 + \cdots + n^3 + (n+1)^3 &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\ &= (n+1)^2\left(\frac{n^2}{4} + n + 1\right) \\ &= (n+1)^2\left(\frac{n^2+4n+4}{4}\right) \\ &= \frac{(n+1)^2(n+2)^2}{4}. \end{aligned}$$

Thus the statement is true for $n + 1$.

Therefore, by induction, $1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$ for every positive integer n.

2. Let a_n be defined recursively by $a_1 = 1$ and $a_n = \sqrt{1 + a_{n-1}}$. Prove that for all positive integers n, $a_n < 2$.

Solution.

We prove the statement by induction on n.

Base case: We have $a_1 = 1 < 2$. *Inductive step:* Suppose that $a_n < 2$. Then $a_{n+1} = \sqrt{1 + a_n} < \sqrt{3}$. Since $\sqrt{3} < 2$, we have $a_{n+1} < 2$. Therefore, by induction, $a_n < 2$ for every positive integer n.

3. Let $F_1, F_2, F_3, \ldots = 1, 1, 2, 3, 5, 8, \ldots$ be the Fibonacci sequence. Prove that

$$F_1^2 + \cdots + F_n^2 = F_n F_{n+1}.$$

Solution.

Recall that the Fibonacci sequence satisfies $F_{n+2} = F_{n+1} + F_n$. We prove the statement by induction on n.

Base case: When $n = 1$, $F_1^2 = 1$ and $F_1 F_2 = 1 \cdot 1 = 1$. Thus the statement is true for $n = 1$.

Inductive step: Suppose that $F_1^2 + \cdots + F_n^2 = F_n F_{n+1}$. Then

$$\begin{aligned} F_1^2 + \cdots + F_n^2 + F_{n+1}^2 &= F_n F_{n+1} + F_{n+1}^2 \\ &= F_{n+1}(F_n + F_{n+1}) \\ &= F_{n+1} F_{n+2}. \end{aligned}$$

Thus the statement is true for $n + 1$.

Therefore, by induction, $F_1^2 + \cdots + F_n^2 = F_n F_{n+1}$ for every positive integer n.

4. Prove that $n! > 2^n$ for all $n \geq 4$.

   Solution.

   We prove the statement by induction on n.

   Base case: When $n = 4$, $4! = 24 > 16 = 2^4$. *Inductive step:* Suppose that for some $n \geq 4$, $n! > 2^n$. Then $(n+1)! = (n+1)n! > (n+1)2^n$. Since $n \geq 4$, we have $n + 1 > 2$, and hence $(n+1)2^n > 2 \cdot 2^n = 2^{n+1}$. Therefore $(n+1)! > 2^{n+1}$. Thus, by induction, $n! > 2^n$ for every integer $n \geq 4$.

5. Prove (using induction) that for all integers $n \geq 1$, $2^{2n} - 1$ is divisible by 3.

   Solution.

   We prove the statement by induction on n.

   Base case: When $n = 1$, $2^{2\cdot 1} - 1 = 4 - 1 = 3$, which is divisible by 3.

   Inductive step: Suppose that $3 \mid 2^{2n} - 1$. Thus there exists an integer k such that $2^{2n} - 1 = 3k$. Then

   $$\begin{aligned} 2^{2(n+1)} - 1 &= 4 \cdot 2^{2n} - 1 \\ &= 4(2^{2n} - 1) + 3 \\ &= 4(3k) + 3 \\ &= 3(4k + 1). \end{aligned}$$

   Since $4k + 1$ is an integer, $3 \mid 2^{2(n+1)} - 1$. Therefore, by induction, $3 \mid 2^{2n} - 1$ for every integer $n \geq 1$.

6. *Bernoulli's inequality*: let $\beta \in \mathbb{R}$ be a real number such that $\beta > -1$ and $\beta \neq 0$. Prove that for all integers $n \geq 2$,

   $$(1 + \beta)^n > 1 + n\beta.$$

   Solution.

   We prove the statement by induction on n.

   Base case: When $n = 2$, $(1 + \beta)^2 = 1 + 2\beta + \beta^2$. Since $\beta \neq 0$, we have $\beta^2 > 0$. Therefore $(1 + \beta)^2 > 1 + 2\beta$. *Inductive step:* Suppose that for some $n \geq 2$, $(1 + \beta)^n > 1 + n\beta$. Since $\beta > -1$, we have $1 + \beta > 0$. We may therefore multiply both sides of the inductive hypothesis by $1 + \beta$ without reversing the inequality:

   $$\begin{aligned} (1 + \beta)^{n+1} &> (1 + n\beta)(1 + \beta) \\ &= 1 + (n + 1)\beta + n\beta^2. \end{aligned}$$

   Since $n > 0$ and $\beta \neq 0$, $n\beta^2 > 0$. Therefore $1 + (n+1)\beta + n\beta^2 > 1 + (n+1)\beta$. Thus $(1 + \beta)^{n+1} > 1 + (n+1)\beta$. Therefore, by induction, $(1 + \beta)^n > 1 + n\beta$ for every integer $n \geq 2$.

7. Prove that for all integers $n \geq 1$,

   $$1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}.$$

   Solution.

   We prove the statement by induction on n.

   Base case: When $n = 1$, $1 = \sqrt{1}$. Thus the statement is true.

   Inductive step: Suppose that $1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$. Then

   $$1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n} + \frac{1}{\sqrt{n+1}}.$$

We claim that $\sqrt{n} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$. Indeed,

$$\begin{aligned}\sqrt{n+1} - \sqrt{n} &= \frac{1}{\sqrt{n+1} + \sqrt{n}} \\ &\leq \frac{1}{\sqrt{n+1}}.\end{aligned}$$

Therefore $\sqrt{n} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$. It follows that $1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$. Therefore, by induction, $1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$ for every integer $n \geq 1$.

8. We prove by induction that

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|.$$

For $n = 1$, we have $|x_1| \leq |x_1|$, so the result is true.

For $n = 2$, the statement is exactly the triangle inequality:

$$|x_1 + x_2| \leq |x_1| + |x_2|.$$

Now suppose that for some $k \geq 2$,

$$|x_1 + \cdots + x_k| \leq |x_1| + \cdots + |x_k|.$$

We want to prove the result for $k + 1$. By the triangle inequality,

$$\begin{aligned}|x_1 + \cdots + x_k + x_{k+1}| &= |(x_1 + \cdots + x_k) + x_{k+1}| \\ &\leq |x_1 + \cdots + x_k| + |x_{k+1}| \\ &\leq |x_1| + \cdots + |x_k| + |x_{k+1}|,\end{aligned}$$

where the first inequality is the triangle inequality, and the last inequality follows from the inductive hypothesis.

Therefore, by induction,

$$\boxed{|x_1 + x_2 + \cdots + x_n| \leq |x_1| + \cdots + |x_n|}$$

for every positive integer n.

Solutions 6: Set theory. Basic operations. Proofs with sets

1. Let $A = \{0, 1, 2\}$. Let $B = \{1, \{2\}, 3\}$. Which of the following statements are true?

   Solution.

(a) $0 \in A$	**True**	(e) $2 \in B$	**False**	(i) $2 \in A \cap B$	**False**
(b) $\{0\} \subseteq A$	**True**	(f) $\{2\} \in B$	**True**	(j) $2 \in A - B$	**True**
(c) $\{0\} \in A$	**False**	(g) $\{2\} \subseteq B$	**False**		
(d) $A = B$	**False**	(h) $2 \in A \cup B$	**True**		

2. Let

   $$A = \{n \in \mathbb{Z} \mid n \text{ is a multiple of } 4\}$$

   and

   $$B = \{n \in \mathbb{Z} \mid n^2 \text{ is a multiple of } 4\}.$$

   (a) Prove or disprove: $A \subseteq B$.

   Solution.

   The statement is true.

   Suppose that $n \in A$. Then n is a multiple of 4, so there exists an integer k such that $n = 4k$. Therefore $n^2 = (4k)^2 = 16k^2 = 4(4k^2)$. Since $4k^2$ is an integer, n^2 is a multiple of 4. Hence $n \in B$. Therefore, $A \subseteq B$.

   (b) Prove or disprove: $B \subseteq A$.

   Solution.

   The statement is false.

   For example, $2 \in B$, since $2^2 = 4$ is a multiple of 4.

   However, $2 \notin A$, since 2 is not a multiple of 4.

   Therefore, $B \nsubseteq A$.

3. Recall that

   $$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

   Prove or disprove each of the following:

   (a) $(-1, 1) \subseteq (-2, 2)$.

   Solution.

   The statement is true.

   Suppose that $x \in (-1, 1)$. Then $-1 < x < 1$. Since $-2 < -1$ and $1 < 2$, we have $-2 < x < 2$. Thus $x \in (-2, 2)$. Therefore, $(-1, 1) \subseteq (-2, 2)$.

   (b) $(-1, 2) \subseteq (-2, 1)$.

   Solution.

   The statement is false.

   For example, $\frac{3}{2} \in (-1, 2)$, but $\frac{3}{2} \notin (-2, 1)$. Therefore, $(-1, 2) \nsubseteq (-2, 1)$.

4. Prove that
$$(-10, 5] \cap [0, 10] = [0, 5].$$

Solution.

We prove the two set containments.

First, suppose that $x \in (-10, 5] \cap [0, 10]$. Then $x \in (-10, 5]$ and $x \in [0, 10]$. Thus $x \leq 5$ and $x \geq 0$. Therefore $0 \leq x \leq 5$, so $x \in [0, 5]$. Hence $(-10, 5] \cap [0, 10] \subseteq [0, 5]$. Conversely, suppose that $x \in [0, 5]$. Then $0 \leq x \leq 5$. In particular, $-10 < x \leq 5$, so $x \in (-10, 5]$. Also, $0 \leq x \leq 10$, so $x \in [0, 10]$. Therefore $x \in (-10, 5] \cap [0, 10]$. Hence $[0, 5] \subseteq (-10, 5] \cap [0, 10]$. Thus $(-10, 5] \cap [0, 10] = [0, 5]$.

5. Prove that if $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq C$.

Solution.

Suppose that $A \not\subseteq C$. Then there exists an element x such that $x \in A$ and $x \notin C$. There are two cases.

If $x \notin B$, then $x \in A$ but $x \notin B$, so $A \not\subseteq B$. If $x \in B$, then $x \in B$ but $x \notin C$, so $B \not\subseteq C$. Thus in either case, $A \not\subseteq B$ or $B \not\subseteq C$.

6. Let A, B be sets. Prove each of the following:

   (a) $A \cap B \subseteq A$.

   Solution.

   Suppose that $x \in A \cap B$. By the definition of intersection, $x \in A$ and $x \in B$. In particular, $x \in A$. Therefore, $A \cap B \subseteq A$.

   (b) $A \cap \emptyset = \emptyset$.

   Solution.

   We prove the two set containments.

   First, suppose that $x \in A \cap \emptyset$. Then, by the definition of intersection, $x \in A$ and $x \in \emptyset$. But the empty set has no elements, so this is impossible. Therefore $A \cap \emptyset$ has no elements, and hence $A \cap \emptyset \subseteq \emptyset$. Also, $\emptyset \subseteq A \cap \emptyset$, since the empty set is a subset of every set.

   Therefore, $A \cap \emptyset = \emptyset$.

7. Prove that
$$A \cup (A \cap B) = A.$$

Solution.

We prove the two set containments.

First, suppose that $x \in A \cup (A \cap B)$. Then either $x \in A$ or $x \in A \cap B$. If $x \in A$, then clearly $x \in A$. If $x \in A \cap B$, then by the definition of intersection, $x \in A$. Thus in either case, $x \in A$. Therefore, $A \cup (A \cap B) \subseteq A$. Conversely, suppose that $x \in A$. Then, by the definition of union, $x \in A \cup (A \cap B)$. Therefore, $A \subseteq A \cup (A \cap B)$. Hence $A \cup (A \cap B) = A$.

8. Let n and m be integers. Prove that if m divides n, then
$$n\mathbb{Z} \subseteq m\mathbb{Z}.$$

Solution.

Suppose that $m \mid n$. Then there exists an integer k such that $n = mk$. Now suppose that $x \in n\mathbb{Z}$. By the definition of $n\mathbb{Z}$, there exists an integer ℓ such that $x = n\ell$. Since $n = mk$, we have $x = (mk)\ell = m(k\ell)$. Since $k\ell$ is an integer, $x \in m\mathbb{Z}$. Therefore, $n\mathbb{Z} \subseteq m\mathbb{Z}$.

Solutions 7: Set theory. Basic operations. Proofs with sets

1. Let A, B, and C be sets.

   (a) Suppose $A \subseteq C$ and $B \subseteq C$. We prove that $A \cup B \subseteq C$.
   Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. If $x \in A$, then $x \in C$ because $A \subseteq C$. If $x \in B$, then $x \in C$ because $B \subseteq C$. In either case, $x \in C$. Therefore

   $$\boxed{A \cup B \subseteq C}.$$

   (b) The contrapositive is

   $$\boxed{A \cup B \not\subseteq C \implies (A \not\subseteq C) \vee (B \not\subseteq C).}$$

   (c) The converse is

   $$A \cup B \subseteq C \implies (A \subseteq C) \wedge (B \subseteq C).$$

   This is **true**. Suppose $A \cup B \subseteq C$. If $x \in A$, then $x \in A \cup B$, so $x \in C$. Hence $A \subseteq C$. Similarly, if $x \in B$, then $x \in A \cup B$, so $x \in C$. Hence $B \subseteq C$. Therefore

   $$\boxed{A \subseteq C \text{ and } B \subseteq C}.$$

2. Let A, B, and C be sets.

   (a) The statement is **false**. Take

   $$A = \{1\}, \qquad B = \{2\}, \qquad C = \{1\}.$$

   Then $A \not\subseteq B$ and $B \not\subseteq C$, but $A \subseteq C$. Thus it is not true that $A \not\subseteq C$.

   (b) The statement is **false**. Take

   $$A = \{1\}, \qquad B = \{1, 2\}, \qquad C = \{1\}.$$

   Then $A \subseteq B$ and $B \not\subseteq C$, but $A \subseteq C$. Thus the conclusion $A \not\subseteq C$ is false.

3. Let A, B, C be sets.

   (a) Suppose $B \subseteq C$. We prove that $A - C \subseteq A - B$.
   Let $x \in A - C$. Then $x \in A$ and $x \notin C$. Since $B \subseteq C$, we cannot have $x \in B$, because that would imply $x \in C$. Hence $x \notin B$. Therefore $x \in A - B$, and so

   $$\boxed{A - C \subseteq A - B}.$$

   (b) We prove both directions.
   Suppose first that $A \subseteq B$. If $x \in A$, then $x \in B$ as well, so $x \in A \cap B$. Thus $A \subseteq A \cap B$. Since always $A \cap B \subseteq A$, we have

   $$A \cap B = A.$$

   Conversely, suppose $A \cap B = A$. Let $x \in A$. Since $A = A \cap B$, we have $x \in A \cap B$, and hence $x \in B$. Therefore $A \subseteq B$.
   Thus

   $$\boxed{A \subseteq B \iff A \cap B = A}.$$

4. Let A, B, and C be sets.

   (a) The statement is **false**. For example, let

   $$A = \{1\}, \qquad B = \emptyset, \qquad C = \{2\}.$$

   Then

   $$(A \cap B) \cup C = \emptyset \cup \{2\} = \{2\},$$

   while

   $$A \cap (B \cup C) = \{1\} \cap \{2\} = \emptyset.$$

   Hence the two sets are not equal.

   (b) The statement is **true**. We prove equality by showing both inclusions.
   Let $x \in (A \cap B) \cup C$. Then either $x \in A \cap B$ or $x \in C$. If $x \in A \cap B$, then $x \in A \cup C$ and $x \in B \cup C$. If $x \in C$, then again $x \in A \cup C$ and $x \in B \cup C$. Thus

   $$(A \cap B) \cup C \subseteq (A \cup C) \cap (B \cup C).$$

   Conversely, let $x \in (A \cup C) \cap (B \cup C)$. Then $x \in A \cup C$ and $x \in B \cup C$. If $x \in C$, then $x \in (A \cap B) \cup C$. If $x \notin C$, then the first membership forces $x \in A$ and the second forces $x \in B$, so $x \in A \cap B$. Hence again $x \in (A \cap B) \cup C$. Therefore

   $$\boxed{(A \cap B) \cup C = (A \cup C) \cap (B \cup C)}.$$

   (c) The statement is **true**. Let x be arbitrary. Then

   $$\begin{aligned} x \in (A \cup B) \cap \overline{A} &\iff (x \in A \text{ or } x \in B) \text{ and } x \notin A \\ &\iff x \in B \text{ and } x \notin A \\ &\iff x \in B - A. \end{aligned}$$

   Therefore

   $$\boxed{(A \cup B) \cap \overline{A} = B - A}.$$

5. We prove

   $$c \in (a-b, a+b) \iff |a-c| < b.$$

   If $b \leq 0$, then $(a-b, a+b)$ is empty, and $|a-c| < b$ is impossible because $|a-c| \geq 0$. Thus both statements are false.

   Now suppose $b > 0$. Then

   $$\begin{aligned} c \in (a-b, a+b) &\iff a - b < c < a + b \\ &\iff -b < c - a < b \\ &\iff |c-a| < b \\ &\iff |a-c| < b. \end{aligned}$$

   Hence

   $$\boxed{c \in (a-b, a+b) \iff |a-c| < b}.$$

6. Suppose $A \subseteq B$. We prove that $P(A) \subseteq P(B)$.

   Let $X \in P(A)$. Then $X \subseteq A$. Since $A \subseteq B$, transitivity of inclusion gives $X \subseteq B$. Hence $X \in P(B)$. Therefore

   $$\boxed{P(A) \subseteq P(B)}.$$

   The converse is

   $$P(A) \subseteq P(B) \implies A \subseteq B.$$

   This is also **true**. Since $A \subseteq A$, we have $A \in P(A)$. Therefore $A \in P(B)$, which means precisely that $A \subseteq B$.

7. Let $A = \{0, 1, 2\}$. Then

   $$P(A) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

   Thus:

   (a) $0 \in P(A)$: **False**.
   (b) $\{0\} \in P(A)$: **True**.
   (c) $\{0\} \subseteq P(A)$: **False**.
   (d) $\{1, 2\} \in P(A)$: **True**.
   (e) $\{1, \{1\}\} \subseteq P(A)$: **False**.
   (f) $\{\{0, 1\}, \{1\}\} \subseteq P(A)$: **True**.
   (g) $\emptyset \in P(A)$: **True**.
   (h) $\emptyset \subseteq P(A)$: **True**.
   (i) $\{\emptyset\} \in P(A)$: **False**.
   (j) $\{\emptyset\} \subseteq P(A)$: **True**.

8. (a) For $i \geq 2$, let

   $$A_i = (1/i, i].$$

   The sets are increasing:

   $$A_2 \subseteq A_3 \subseteq A_4 \subseteq \cdots.$$

   Hence their intersection is the smallest set:

   $$\boxed{\bigcap_{i=2}^{\infty} A_i = A_2 = (1/2, 2]}.$$

   We claim that

   $$\boxed{\bigcup_{i=2}^{\infty} A_i = (0, \infty)}.$$

   Every A_i consists only of positive numbers, so the union is contained in $(0, \infty)$. Conversely, let $x > 0$. Choose an integer $i \geq 2$ large enough that

   $$i \geq x \qquad \text{and} \qquad i > 1/x.$$

   Then $1/i < x \leq i$, so $x \in A_i$. Thus every positive real number belongs to the union.

   (b) Let

   $$A_i = [1, 1 + 1/i], \qquad i \geq 1.$$

   These sets are decreasing:

   $$A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots.$$

   Since $A_1 = [1, 2]$, every A_i is contained in $[1, 2]$, and A_1 itself occurs in the union. Therefore

   $$\boxed{\bigcup_{i=1}^{\infty} A_i = [1, 2]}.$$

Also, $1 \in A_i$ for every i, so

$$\{1\} \subseteq \bigcap_{i=1}^{\infty} A_i.$$

Now let $x \neq 1$. If $x < 1$, then $x \notin A_1$. If $x > 1$, choose i large enough that

$$\frac{1}{i} < x - 1.$$

Then $1 + 1/i < x$, so $x \notin A_i$. Thus no number other than 1 belongs to every A_i. Hence

$$\boxed{\bigcap_{i=1}^{\infty} A_i = \{1\}}.$$

Solutions 8: Introduction to functions; images and surjectivity

1. Let $f\colon A \to B$ be a function. Finish the following sentence: an element $b \in B$ is not in the image of f if $\cdots$ **Solution.**

   An element $b \in B$ is not in the image of f if for every $a \in A$, $f(a) \neq b$. Equivalently, there does not exist an $a \in A$ such that $f(a) = b$.

2. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function defined by $f(x) = 6x + 5$.

   (a) Prove that $f(\mathbf{R}) = \mathbf{R}$.

   Solution.

   We already know that $f(\mathbf{R}) \subseteq \mathbf{R}$ because the codomain of f is $\mathbf{R}$.

   To prove the other containment, let $y \in \mathbf{R}$. Set $x = \frac{y-5}{6}$. Then $x \in \mathbf{R}$, and $f(x) = 6\left(\frac{y-5}{6}\right) + 5 = y$. Thus $y \in f(\mathbf{R})$. Therefore $\mathbf{R} \subseteq f(\mathbf{R})$. Hence $f(\mathbf{R}) = \mathbf{R}$.

   (b) Compute $f([1, 4])$. Prove your answer.

   Solution.

   We claim that $f([1, 4]) = [11, 29]$. First suppose that $y \in f([1, 4])$. Then there exists some $x \in [1, 4]$ such that $y = f(x) = 6x + 5$. Since $1 \leq x \leq 4$, we have $6 \leq 6x \leq 24$, and hence $11 \leq 6x + 5 \leq 29$. Thus $y \in [11, 29]$. Therefore $f([1, 4]) \subseteq [11, 29]$. Conversely, suppose that $y \in [11, 29]$. Set $x = \frac{y-5}{6}$. Since $11 \leq y \leq 29$, we have $1 \leq \frac{y-5}{6} \leq 4$. Thus $x \in [1, 4]$. Moreover, $f(x) = 6\left(\frac{y-5}{6}\right) + 5 = y$. Hence $y \in f([1, 4])$. Therefore $f([1, 4]) = [11, 29]$.

3. Let g be the function $g\colon \mathbf{Z} \to \mathbf{Z}$, where

$$g(x) = 2x + 1.$$

   (a) Compute the image of g. Prove that your answer is correct.

   Solution.

   We claim that the image of g is the set of odd integers: $g(\mathbf{Z}) = \{2k + 1 : k \in \mathbf{Z}\}$. First, suppose that $y \in g(\mathbf{Z})$. Then there exists an integer x such that $y = g(x) = 2x + 1$. Therefore y is odd.

   Conversely, suppose that y is an odd integer. Then, by the definition of odd, there exists an integer k such that $y = 2k + 1$. But $g(k) = 2k + 1 = y$. Thus $y \in g(\mathbf{Z})$. Therefore the image of g is precisely the set of odd integers.

4. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function defined by

$$f(x) = x^4 + x^2.$$

   (a) Compute the image of f. Prove that your answer is correct.

   Solution.

   We claim that $f(\mathbf{R}) = [0, \infty)$. For every real number x, $x^4 \geq 0$ and $x^2 \geq 0$. Therefore $f(x) = x^4 + x^2 \geq 0$. Thus $f(\mathbf{R}) \subseteq [0, \infty)$. Conversely, let $y \in [0, \infty)$. We want to find a real number x such that $x^4 + x^2 = y$. Let $t = \frac{-1+\sqrt{1+4y}}{2}$. Since $y \geq 0$, we have $t \geq 0$. Also, $t^2 + t = y$. Choose $x = \sqrt{t}$. Then $x^2 = t$, and hence $f(x) = x^4 + x^2 = t^2 + t = y$. Thus $y \in f(\mathbf{R})$. Therefore $f(\mathbf{R}) = [0, \infty)$.

(b) Compute $f([-1,2])$. Prove that your answer is correct.

Solution.

We claim that $f([-1,2]) = [0,20]$. If $x \in [-1,2]$, then $0 \leq x^2 \leq 4$. Let $t = x^2$. Then $f(x) = t^2 + t$. For $0 \leq t \leq 4$, we have $0 \leq t^2 + t \leq 4^2 + 4 = 20$. Thus $f([-1,2]) \subseteq [0,20]$. Conversely, let $y \in [0,20]$. Set $t = \frac{-1+\sqrt{1+4y}}{2}$. Since $0 \leq y \leq 20$, we have $0 \leq t \leq 4$. Set $x = \sqrt{t}$. Then $0 \leq x \leq 2$, so $x \in [-1,2]$. Moreover, $f(x) = t^2 + t = y$. Thus $y \in f([-1,2])$. Therefore $f([-1,2]) = [0,20]$.

5. Let $g\colon \mathbb{R} \to \mathbb{Z}$ be the **ceiling function**

$$g(x) = \lceil x \rceil.$$

Compute the image of g. Prove that your answer is correct.

Solution.

We claim that $g(\mathbf{R}) = \mathbf{Z}$. Since the codomain of g is $\mathbf{Z}$, $g(\mathbf{R}) \subseteq \mathbf{Z}$. Conversely, suppose that $n \in \mathbf{Z}$. Since n is already an integer, $\lceil n \rceil = n$. Thus $g(n) = n$, so $n \in g(\mathbf{R})$. Therefore $\mathbf{Z} \subseteq g(\mathbf{R})$. Hence $g(\mathbf{R}) = \mathbf{Z}$.

6. Consider the function $\sin\colon \mathbf{R} \to \mathbf{R}$.

(a) Compute the image of sin. Prove that your answer is correct.

Solution.

We claim that $\sin(\mathbf{R}) = [-1,1]$. For every real number x, $-1 \leq \sin x \leq 1$. Therefore $\sin(\mathbf{R}) \subseteq [-1,1]$. Conversely, every real number between -1 and 1 occurs as the sine of some real number. In particular, if $y \in [-1,1]$, then $x = \arcsin(y)$ is a real number and $\sin x = y$. Thus $[-1,1] \subseteq \sin(\mathbf{R})$. Therefore $\sin(\mathbf{R}) = [-1,1]$.

(b) Compute $\sin([0,\pi/4])$. Prove that your answer is correct.

Solution.

We claim that $\sin([0,\pi/4]) = \left[0, \frac{\sqrt{2}}{2}\right]$. The sine function is increasing on $[0,\pi/2]$. Therefore, if $0 \leq x \leq \frac{\pi}{4}$, then $\sin 0 \leq \sin x \leq \sin\left(\frac{\pi}{4}\right)$. Since $\sin 0 = 0$ and $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$, we have $\sin([0,\pi/4]) \subseteq \left[0, \frac{\sqrt{2}}{2}\right]$. Conversely, suppose that $y \in \left[0, \frac{\sqrt{2}}{2}\right]$. Then $x = \arcsin(y)$ satisfies $0 \leq x \leq \frac{\pi}{4}$ and $\sin x = y$. Thus $y \in \sin([0,\pi/4])$. Therefore $\sin([0,\pi/4]) = \left[0, \frac{\sqrt{2}}{2}\right]$.

7. Let A and B be sets and let X and Y be subsets of A. Let $f\colon A \to B$ be a function. Prove or disprove each of the following.

(a)

$$f(X \cup Y) \subseteq f(X) \cup f(Y).$$

Solution.

The statement is true.

Suppose that $b \in f(X \cup Y)$. Then there exists some $x \in X \cup Y$ such that $f(x) = b$. Since $x \in X \cup Y$, either $x \in X$ or $x \in Y$. If $x \in X$, then $b = f(x) \in f(X)$. If $x \in Y$, then $b = f(x) \in f(Y)$. Thus in either case, $b \in f(X) \cup f(Y)$. Therefore, $f(X \cup Y) \subseteq f(X) \cup f(Y)$.

(b)

$$f(X \cup Y) \supseteq f(X) \cup f(Y).$$

Solution.

The statement is also true.

Suppose that $b \in f(X) \cup f(Y)$. Then either $b \in f(X)$ or $b \in f(Y)$. If $b \in f(X)$, then there exists some $x \in X$ such that $f(x) = b$. Since $X \subseteq X \cup Y$, we have $x \in X \cup Y$, and therefore $b \in f(X \cup Y)$. The case $b \in f(Y)$ is similar.

Therefore, $f(X) \cup f(Y) \subseteq f(X \cup Y)$. In particular, the two parts together show that $f(X \cup Y) = f(X) \cup f(Y)$.

8. Let A and B be sets and let X and Y be subsets of A. Let $f\colon A \to B$ be a function. Prove or disprove each of the following.

(a)
$$f(X) - f(Y) \subseteq f(X - Y).$$

Solution.

The statement is true.

Suppose that $b \in f(X) - f(Y)$. Then $b \in f(X)$ and $b \notin f(Y)$. Since $b \in f(X)$, there exists some $x \in X$ such that $f(x) = b$. We claim that $x \notin Y$. If $x \in Y$, then $b = f(x) \in f(Y)$, which contradicts the fact that $b \notin f(Y)$. Therefore $x \in X - Y$. Since $f(x) = b$, we have $b \in f(X - Y)$. Therefore, $f(X) - f(Y) \subseteq f(X - Y)$.

(b)
$$f(X) - f(Y) \supseteq f(X - Y).$$

Solution.

The statement is false.

For example, let $A = \{1, 2\}$, $B = \{0\}$, and define $f(1) = f(2) = 0$. Let $X = \{1\}$, $Y = \{2\}$. Then $X - Y = \{1\}$, so $f(X - Y) = \{0\}$. On the other hand, $f(X) = \{0\}$ and $f(Y) = \{0\}$. Therefore $f(X) - f(Y) = \{0\} - \{0\} = \emptyset$. Hence $f(X) - f(Y) \not\supseteq f(X - Y)$. Therefore the statement is false.

Solutions 9: Inverse Image (or Preimage)

1. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function defined by $f(x) = 3x + 1$.

   (a) Compute $f^{-1}(1, 5, 8)$.
   Solution. $f^{-1}(\{1, 5, 8\}) = \left\{0, \frac{4}{3}, \frac{7}{3}\right\}$.

   (b) Compute $f^{-1}(W)$, where $W = (4, \infty)$, and give a proof that your answer is correct.
   Solution.
   We claim that $f^{-1}(W) = (1, \infty)$. Suppose that $x \in f^{-1}(W)$. Then $f(x) \in W$, so $3x + 1 > 4$. Thus $x > 1$, and therefore $x \in (1, \infty)$. Hence $f^{-1}(W) \subseteq (1, \infty)$. Conversely, suppose that $x \in (1, \infty)$. Then $x > 1$, so $3x + 1 > 4$. Thus $f(x) \in (4, \infty) = W$, and therefore $x \in f^{-1}(W)$. Hence $f^{-1}(W) = (1, \infty)$.

   (c) Compute $f^{-1}(\mathbf{E})$, where $\mathbf{E}$ is the set of even integers, and give a proof that your answer is correct.
   Solution.
   We claim that $f^{-1}(\mathbf{E}) = \left\{\frac{2k-1}{3} : k \in \mathbf{Z}\right\}$. Suppose that $x \in f^{-1}(\mathbf{E})$. Then $f(x)$ is an even integer. Thus there exists an integer k such that $3x + 1 = 2k$. Solving for x gives $x = \frac{2k-1}{3}$. Therefore, $x \in \left\{\frac{2k-1}{3} : k \in \mathbf{Z}\right\}$. Conversely, suppose that $x = \frac{2k-1}{3}$ for some integer k. Then $f(x) = 3\left(\frac{2k-1}{3}\right) + 1 = 2k$. Thus $f(x)$ is an even integer, so $x \in f^{-1}(\mathbf{E})$. Therefore, $f^{-1}(\mathbf{E}) = \left\{\frac{2k-1}{3} : k \in \mathbf{Z}\right\}$.

2. Let $f\colon \mathbf{Z} \to \mathbf{Z}$ be the function defined by

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even,} \\ 2n + 4, & \text{if } n \text{ is odd.} \end{cases}$$

   Compute $f^{-1}(\mathbf{E})$. Prove that your answer is correct.

   Solution.

   We claim that $f^{-1}(\mathbf{E}) = \{n \in \mathbf{Z} : n \text{ is odd}\} \cup 4\mathbf{Z}$. First suppose that $n \in f^{-1}(\mathbf{E})$. Thus $f(n)$ is even.

   If n is odd, then $n \in \{n \in \mathbf{Z} : n \text{ is odd}\}$. If n is even, then $f(n) = \frac{n}{2}$. Since $f(n)$ is even, there exists an integer k such that $\frac{n}{2} = 2k$. Therefore $n = 4k$, so $n \in 4\mathbf{Z}$. Thus $f^{-1}(\mathbf{E}) \subseteq \{n \in \mathbf{Z} : n \text{ is odd}\} \cup 4\mathbf{Z}$. Conversely, suppose first that n is odd. Then $f(n) = 2n + 4 = 2(n + 2)$, which is even. Thus $n \in f^{-1}(\mathbf{E})$. Now suppose that $n \in 4\mathbf{Z}$. Then $n = 4k$ for some integer k. Since n is even, $f(n) = \frac{n}{2} = 2k$, which is even. Thus again $n \in f^{-1}(\mathbf{E})$. Therefore, $f^{-1}(\mathbf{E}) = \{n \in \mathbf{Z} : n \text{ is odd}\} \cup 4\mathbf{Z}$.

3. Let A and B be sets and let X and Y be subsets of B. Let $f\colon A \to B$ be a function. Prove or disprove the following.

   (a)
$$f^{-1}(X \cap Y) \subseteq f^{-1}(X) \cap f^{-1}(Y).$$

   Solution.

   The statement is true.

   Suppose that $a \in f^{-1}(X \cap Y)$. Then $f(a) \in X \cap Y$. Thus $f(a) \in X$ and $f(a) \in Y$. Therefore $a \in f^{-1}(X)$ and $a \in f^{-1}(Y)$, so $a \in f^{-1}(X) \cap f^{-1}(Y)$. Hence $f^{-1}(X \cap Y) \subseteq f^{-1}(X) \cap f^{-1}(Y)$.

(b)

$$f^{-1}(X \cap Y) \supseteq f^{-1}(X) \cap f^{-1}(Y).$$

Solution.

The statement is also true.

Suppose that $a \in f^{-1}(X) \cap f^{-1}(Y)$. Then $a \in f^{-1}(X)$ and $a \in f^{-1}(Y)$. Thus $f(a) \in X$ and $f(a) \in Y$. Therefore $f(a) \in X \cap Y$, so $a \in f^{-1}(X \cap Y)$. Hence $f^{-1}(X) \cap f^{-1}(Y) \subseteq f^{-1}(X \cap Y)$. Thus the two parts together show that $f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$.

4. Let A and B be sets and let X be a subset of B. Let $f\colon A \to B$ be a function. Prove or disprove the following.

(a)

$$X \subseteq f(f^{-1}(X)).$$

Solution.

The statement is false.

For example, let $A = \{1\}$, $B = \{a, b\}$, and define $f(1) = a$. Let $X = \{b\}$. Since no element of A maps to b, $f^{-1}(X) = \emptyset$. Therefore $f(f^{-1}(X)) = f(\emptyset) = \emptyset$. But $X = \{b\} \not\subseteq \emptyset$. Therefore the statement is false.

(b)

$$X \supseteq f(f^{-1}(X)).$$

Solution.

The statement is true.

Suppose that $b \in f(f^{-1}(X))$. Then there exists some $a \in f^{-1}(X)$ such that $f(a) = b$. Since $a \in f^{-1}(X)$, we have $f(a) \in X$. Therefore $b \in X$. Thus $f(f^{-1}(X)) \subseteq X$, or equivalently, $X \supseteq f(f^{-1}(X))$.

5. Let A and B be sets. Let $S \subseteq A$ and let $T \subseteq B$. Let $f\colon A \to B$ be a function. Prove or disprove the following.

(a)

$$f(S) \subseteq T \Rightarrow S \subseteq f^{-1}(T).$$

Solution.

The statement is true.

Suppose that $f(S) \subseteq T$. Let $x \in S$. Then $f(x) \in f(S)$. Since $f(S) \subseteq T$, we have $f(x) \in T$. Therefore $x \in f^{-1}(T)$. Thus $S \subseteq f^{-1}(T)$.

(b)

$$S \subseteq f^{-1}(T) \Rightarrow f(S) \subseteq T.$$

Solution.

The statement is true.

Suppose that $S \subseteq f^{-1}(T)$. Let $y \in f(S)$. Then there exists some $x \in S$ such that $f(x) = y$. Since $x \in S \subseteq f^{-1}(T)$, we have $f(x) \in T$. Thus $y \in T$. Therefore $f(S) \subseteq T$. Thus, in fact, $f(S) \subseteq T \iff S \subseteq f^{-1}(T)$.

In each of the following problems, let $f\colon A \to A$ be a function and suppose that $C \subseteq A$.

6. Prove or disprove the following.

   (a)
   $$f^{-1}(C) \subseteq C.$$

   Solution.
   The statement is false.
   Let $A = \{1, 2\}$, $C = \{1\}$, and define $f(1) = 2$, $f(2) = 1$. Then $f^{-1}(C) = \{2\}$. Therefore $f^{-1}(C) \not\subseteq C$.

   (b)
   $$C \subseteq f^{-1}(C).$$

   Solution.
   The statement is false.
   Using the same example, $A = \{1, 2\}$, $C = \{1\}$, and $f(1) = 2$, $f(2) = 1$, we have $f^{-1}(C) = \{2\}$. Thus $C = \{1\} \not\subseteq \{2\} = f^{-1}(C)$.

   (c)
   $$f(C) \subseteq C.$$

   Solution.
   The statement is false.
   Again let $A = \{1, 2\}$, $C = \{1\}$, and define $f(1) = 2$, $f(2) = 1$. Then $f(C) = \{2\}$, so $f(C) \not\subseteq C$.

   (d)
   $$C \subseteq f(C).$$

   Solution.
   The statement is false.
   Using the same example, $f(C) = \{2\}$. Thus $C = \{1\} \not\subseteq \{2\} = f(C)$.

7. Prove or disprove the following:

   $$C \subseteq f^{-1}(C) \iff f(C) \subseteq C.$$

   Solution.

   The statement is true.

   First suppose that $C \subseteq f^{-1}(C)$. Let $y \in f(C)$. Then there exists some $x \in C$ such that $f(x) = y$. Since $x \in C \subseteq f^{-1}(C)$, we have $f(x) \in C$. Thus $y \in C$. Therefore $f(C) \subseteq C$. Conversely, suppose that $f(C) \subseteq C$. Let $x \in C$. Then $f(x) \in f(C)$. Since $f(C) \subseteq C$, we have $f(x) \in C$. Therefore $x \in f^{-1}(C)$. Thus $C \subseteq f^{-1}(C)$. Hence $C \subseteq f^{-1}(C) \iff f(C) \subseteq C$.

8. Prove or disprove the following:

   $$f^{-1}(C) \subseteq C \iff C \subseteq f(C).$$

   Solution.

   The statement is false.

   For example, let $A = \{1, 2\}$, $C = \{1\}$, and define $f(1) = 2$, $f(2) = 2$. Then no element of A maps to 1, so $f^{-1}(C) = \emptyset$. Thus $f^{-1}(C) \subseteq C$. On the other hand, $f(C) = \{2\}$. Hence $C = \{1\} \not\subseteq \{2\} = f(C)$. Therefore $f^{-1}(C) \subseteq C$ does not imply $C \subseteq f(C)$, so the two statements are not equivalent.